

STEMvibe
Integral Cup 2026: Grand Prix
Practice Questions

Number Theory

- Q1.** Determine the number of integers n with $1 \leq n \leq 600$ for which $n^2 \equiv n \pmod{30}$.
- Q2.** Prove that the product of four consecutive positive integers is never a perfect square.
- Q3.** A sequence is defined by $a_1 = 2$ and $a_{n+1} = a_n^2 - a_n + 1$ for $n \geq 1$. Prove that $\gcd(a_m, a_n) = 1$ whenever $m \neq n$, and deduce that there are infinitely many primes.
- Q4.** Find all positive integers n for which $n \cdot 2^n + 1$ is a perfect square.
- Q5.** Let $p > 3$ be a prime. Prove that the numerator of the reduced fraction

$$1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{p-1}$$

is divisible by p^2 .

- Q6.** Let a, b be positive integers with $ab > 1$ such that $ab - 1$ divides $a^2 + b^2$. Prove that

$$\frac{a^2 + b^2}{ab - 1} = 5.$$

— *End of paper* —